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23IT014

Worksheet - Elliptic Curve Cryptosystem

1. Identify at least four affine points on the Elliptic Curve $E_{23}(1,1)$.

$$y^2 = x^3 + ax + b$$

$$y^2 = x^3 + x + 1$$

x	y^2	y	$(x, y$
0	1	± 1	$(0,1)$ $(0,-1) \equiv (0,22)$
1	3 49	± 7	$(0,7)$ $(0,-7) \equiv (0,16)$
2	11 → 34, 47	–	–
3	31 → 31 mod 23 → 8. 100	± 10	$(3,10)$ $(3,10) \equiv (3,13)$

affine points on $E_{23}(1,1) \Rightarrow (0,1), (0,22),$
$(0,7)$ $(0,16)$ $(3,10)$, $(3,13)$.

2. In an Elliptic curve key exchange algorithm, the cryptosystem parameters are $E_{23}(9,17)$ and $G(16,5)$

a. If user A has a private key 2, What is his public key

Public key = (x_3, y_3)

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For $P = (x_1, y_1)$ with $P = Q$

$Q = (x_1, y_1)$

$$\lambda = \frac{3x_1^2 + a}{2y_1} \bmod p$$

$$x_3 = \lambda^2 - 2x_1 \quad ; \quad y_3 = \lambda(x_1 - x_3) - y_1$$

Ex $P = (x_1, y_1)$; $Q = (x_2, y_2)$ with $P \neq Q$

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} \pmod p$$

$$x_3 = \lambda^2 - x_1 - x_2 \quad ; \quad y_3 = \lambda(x_1 - x_3) - y_1$$

a) Given private key $k_A = 2$; publickey $P_A = 2G$

$$\lambda = \frac{3 \cdot 16^2 + 9}{2 \times 5} \bmod 23$$

$777 \equiv 18 \bmod 23$

$10^{-1} \equiv 7 \bmod 23$

$$\Rightarrow \frac{777}{10} \bmod 23$$

$$\lambda = 18 \cdot 7 = 11$$

$$x_3 = 11^2 - (2 \cdot 16) = 121 - 32 = 20 \pmod{23} = 20$$

$$y_3 = 11(16 - 20) - 5 = 11(-4) - 5 = 20 \pmod{23} = 20$$

$$\boxed{P_A = 2G = (20, 20)}$$

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b) If user B has a private key 4, what is his public key?

$$4G = 2(2G)$$

From a) $2G = (20, 20)$

$$\lambda = \frac{3 \cdot 20^2 + 9}{2 \times 20} \bmod 23$$

$$= 1209 \cdot 40^{-1} \bmod 23$$

$$= 13 \cdot 17^{-1} \bmod 23$$

Q	A_1	A_2	A_3	B_1	B_2	B_3
	1	0	23	0	4	17
1	0	1	17	1-0 1	0-1 -1	23-17 6
2	1	-1	~~0-1~~ 6	0+2 -2	1+2 3	17-12 5
1	-2	3	5	1+2 3	-1-3 -4	6-5 1

$$-4 + 23 = 19$$

$$\Rightarrow 13 \cdot 19 \bmod 23 = 17$$

$$\boxed{\lambda = 17}$$

$$x_3 = 17^2 - (2 \cdot 20) \equiv 289 - 40 \equiv 19 \pmod{23}$$

$$= 19$$

$$y_3 = 17(20 - 19) - 20) \equiv 17 - 20 \equiv 20 \pmod{23} = 20$$

$$\boxed{P_{A} = (19, 20)}$$

c) what is the shared secret key ?

Shared secret $= K_A P_B = K_B P_A =$

$= 2.(19.20) = 2.(2G)$

$= \cancel{2.(4G) = 2(}$

$= (12,17)$

$\boxed{\text{Shared Secret} = (12,17)}$

3. In an elliptic curve key exchange algorithm, the cryptosystem parameters are $E_{11}(1,6)$ and $G = (2,7)$

Soln:

curve $\Rightarrow y^2 = x^3 + ax + b$

$y^2 = x^3 + x + 6 \mod 11$

To verify point

LHS $\Rightarrow y^2 = 7^2 = 49 \equiv 5 \mod 11 \Rightarrow 5$

RHS $\Rightarrow x^3 + x + 6 \Rightarrow 8 + 2 + 6 \equiv 16 \mod 11 = 5$

LHS = RHS

$\therefore G = (2,7)$ is a valid point on the curve

a. If user has private key 4, what is his public key?

$K_A = 4$; $P_A = ?$

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$2G = G + G$ $\quad a = 1 \quad G = (2, 7)$

$x_1 = 2, \; y_1 = 7$

$$\lambda = \frac{2x_1^2 + a}{2y_1}$$

$$= \frac{3(2)^2 + 1}{2 \times 7} = \frac{13}{14} \bmod 11$$

$$= 2 \times 4 = (8 \bmod 11)$$

$$\lambda = 8$$

$$x_3 = \lambda^2 - 2x_1 = 8^2 - 4 = 60 \equiv 5 \pmod{11} = 5$$

$$y_3 = \lambda(x_1 - x_3) - y_1 = 8(2-5) - 7 = 8(-3) - 7$$

$$= 2$$

$$2G = (5, 2)$$

$3G \Rightarrow \quad 3G = 2G + G$

$2G = (5, 2) \quad G = (2, 7)$

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} \bmod p$$

$$\lambda = \frac{7 - 2}{2 - 5} = \cancel{\frac{5}{-3} = 3 \equiv 8 \bmod 11 \Rightarrow 8}$$

$$= \frac{5}{-3} \bmod 11 \qquad -3 \bmod 11 = 7$$

$$= 5 \times 7 = 35 \bmod 11 \Rightarrow 2$$

$$\boxed{\lambda = 2}$$

$x_3 = 2^2 - 5 - 2 = 4 - 7 = -3 \Rightarrow 8$

$y_3 = 2(5-8) - 2 = 2(-3) - 2 = -6 - 2 = -8 + 11$
$= 3$

$\boxed{3G = (8,3)}$

$4G \Rightarrow 3G + G$

$3G = (8,3) \quad G = (2,7)$

$$\lambda = \frac{7-3}{2-8} = \frac{4}{-6} \bmod 11$$

$\Rightarrow 36 \bmod 11 = 3$

$\lambda = 3$

$x_3 = 3^2 - 8 - 2 = 9 - 10 = -1 \bmod 11 \Rightarrow 10$

$y_3 = 3(10 - 10) - 3) = 3(-2) - 3 = -6 - 3 = -9$

$= 2$

$4G = (10,2)$

$\therefore$ public key $P_A = (10,2)$

b) Encrypt the message $(10,9)$ using random value 3

$G_n \Rightarrow M = (10,9) \; ; k = 3 \; ; P_A = (10,2)$

ciphertext

$C_1 = \overset{3 \times G}{kG} \; ; \; C_2 = M + kP_A$

$3G = (8,3)$

$C_2 = (8,3) \times 3 \times (10,2)$

$2P_A$:

$$2P_A = (10,2) + (10,2)$$

$$\lambda = \frac{3x_1^2 + a}{2y_1} = \frac{3(10)^2 + 1}{2(2)} = \frac{301}{4} \bmod 11$$

so $\lambda = 4 \times 3 = 1 \bmod 11$

$$\lambda = 1$$

$x_3 = 1^2 - 2 \times 10 = 1 - 20 \Rightarrow -19 \bmod 11 \Rightarrow 3$

$y_3 = 1(10-3) - 2 = 7 - 2 = 5 \bmod 11 \Rightarrow 5$

$\therefore 2P_A = (3,5)$

$3P_A$:

$$3P_A = 2P_A + P_A = (3,5) + (10,2)$$

$$\lambda = \frac{2-5}{10-3} = \frac{-3}{7} \Rightarrow \frac{8}{7} \bmod 11$$

$\Rightarrow 8 \times 8 \Rightarrow 64 \equiv 9 \bmod 11$

$$\lambda = 9$$

$x_3 = 9^3 - 3 - 10 = 81 - 13 = 68 \Rightarrow 2 \pmod{11} \Rightarrow 2$

$y_3 = 9(3-2) - 5 = 9 \times (1-5) = 4 \bmod 11 \Rightarrow 4$

$3P_A = (2,4) \quad \therefore K_{PA} = (2,4)$

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$C_2 = M + kP_A$

$M = (10, 9)$; $kP_A = (2, 4)$

$\lambda = \frac{4-9}{2-10} \Rightarrow \frac{-5}{-8} = \frac{6}{3} \bmod 11$

$6 \times 4 = 24 = 2 \bmod 11 = 2$

$\boxed{\lambda = 2}$

$x_3 = 2^2 - 10 - 2 \Rightarrow 3$

$y_3 = 2(10-3) - 9 = 5$

$C_2 = (3, 5)$

$\therefore C = ((8, 3), (3, 5))$ is the ciphertext.

c) Show the calculation by which the plaintext is recovered.

Decryption $M = C_2 - aC_1$

where $C_1 = (8, 3)$ $C_2 = (3, 5)$

$aC_1 = 4 \times (8, 3)$

$2C_1 = (5, 9)$ $3C_1 = (10, 9)$ $4C_1 = (2, 4)$

$\boxed{\therefore aC_1 = (2, 4)}$

$M = (3, 5) - (2, 4) = (3, 5) + (2, -4)$

$M = (3+5) + (2, 7)$

$\lambda = \frac{7-5}{2-3} = \frac{2}{-1} = \frac{2}{10} \bmod 11$

$\lambda = 2 \times 10 \equiv 20 \bmod 11$

$\boxed{\lambda = 9}$

$$x_3 = 9^2 - 3 - 2 \equiv 81 - 5 \equiv 76 \bmod 11$$

$$= 10$$

$$y_3 = 9(3 - 10) - 5 \equiv 9(-7) - 5$$

$$\equiv 63 - 5 \equiv -68 \bmod 11$$

$$\boxed{= 9}$$

$M(10, 9)$ is the recovered msg.